ROBUST POSTERIOR-AMBIGUITY MAPS FOR REDUCED-LEAD ECG RECONSTRUCTION

ABSTRACT

Reduced-lead electrocardiography needs more than a mean reconstruction score: difficulty varies with the observed set, waveform segment, and missing target. We define a model-conditional posterior ambiguity in millivolts under a fitted Gaussian spatial prior. One observation variance is frozen on PTB-XL fold 8, while ranks 2–5 are retained and combined by the worst patient-bootstrap 97.5th percentile. The resulting map is tested as a predictor, not as a clinical guarantee. The primary estimand is the leave-one-configuration-out gain in predicting patient-level log-RMSE when ambiguity is added to prespecified covariates. Fold 9 fits the meta-model, fold 10 is opened once, and zero-transfer tests use both Chapman and CPSC 2018. Four reconstructors share a fixed 64-set panel; ECGrecover is reported separately on its supported single-input task. Numerical claims and figures are released only after the hard Stage 15 rule and human review. PENDING–STAGE 15

***Index Terms*—** electrocardiography, reduced leads, posterior uncertainty, reconstruction, external validation

1. MOTIVATION AND CONTRIBUTION

Reduced-lead systems motivate fixed transforms, regression, and learned completion of the standard 12-lead display [1, 2, 3, 4, 5, 6, 7, 8]. Their reconstruction metrics answer how well a method performed after training. The latest multi-model benchmark exhaustively evaluates 4,094 nominal 12-lead subsets for LR, ridge, and LightCNN, with 32 representative Transformer configurations [9]. It ranks configurations by aggregate fidelity, diagnosis, and electrode burden; it does not identify, before choosing a reconstructor, which missing lead and waveform segment are weakly constrained by an observed-lead set. A global condition number also gives every target the same configuration-level description, despite the algebraic dependence of displayed limb leads and target-specific spatial geometry.

For segment s, nonempty observed set S, and target lead $\ell \notin S$, we test one empirical claim: a training-derived robust ambiguity $A_{\mathrm{robust}}(s, \ell, S)$ adds held-out predictive value for reconstruction error beyond method, segment, target, observed-lead count, configuration rank, conditioning, target scale, and observed–target correlation. The claim must hold on the untouched PTB-XL test fold, cover most common-panel methods, and show positive uncertainty-bounded transfer in at least one of two prespecified external cohorts.

Contributions. First, we give a target-resolved posterior standard deviation in physical units and a conservative rank-and-patient envelope. Second, we freeze a falsifiable benchmark with four common-panel reconstructors, one separate strong single-input baseline, and two zero-transfer cohorts. Third, we connect every claim to a hard automatic decision followed by human review. The current manuscript contains the complete protocol and no invented result.

Sensor-design objectives usually optimize a global acquisition criterion [10]. Our setting is descriptive: S is given, and the proposed quantity must earn its interpretation by predicting unseen reconstruction outcomes. It is therefore narrower than a biological identifiability statement and does not prescribe a lead set.

2. GAUSSIAN POSTERIOR AMBIGUITY

2.1. Frozen spatial model and observation variance

At 500 Hz, QRS, ST, and T samples from PTB-XL folds 1–7 fit one spatial model for each $r \in \mathcal{R} = \{2, 3, 4, 5\}$. The primary representation fits the eight algebraically independent channels $[\mathrm{I}, \mathrm{II}, \mathrm{V1}, \ldots, \mathrm{V6}]$ and lifts them to the displayed 12 leads; raw 12-lead PCA is only a sensitivity analysis. For a centered lead vector,

$$\boldsymbol{x}_s - \boldsymbol{\mu}_s = \boldsymbol{M}_{s,r}\boldsymbol{d}, \qquad \boldsymbol{d} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{\Sigma}_{s,r}), \tag{1}$$

where $\boldsymbol{M}_{s,r} \in \mathbb{R}^{12\times r}$. Observing S is represented as $\boldsymbol{y}_S = \boldsymbol{\mu}_S + \boldsymbol{M}_S\boldsymbol{d} + \boldsymbol{\varepsilon}$ with $\boldsymbol{\varepsilon} \sim \mathcal{N}(\boldsymbol{0}, \tau^2\boldsymbol{I})$. The coordinate posterior covariance is

$$\boldsymbol{\Sigma}^{\mathrm{post}}_{s,r,S} = \boldsymbol{\Sigma} - \boldsymbol{\Sigma}\boldsymbol{M}_S^\top(\boldsymbol{M}_S\boldsymbol{\Sigma}\boldsymbol{M}_S^\top + \tau^2\boldsymbol{I})^{-1}\boldsymbol{M}_S\boldsymbol{\Sigma}. \tag{2}$$

Only one τ^2 is used for all segments, ranks, and configurations. Fold 8 selects it from the prespecified log grid $\{10^{-8}, \ldots, 10^{-1}\}$ mV^2 by mean patient-balanced missing-target MSE, giving equal weight to each segment–rank–configuration cell. The chosen value is then frozen. Fold 8 never selects a rank.

For target row $\boldsymbol{m}_{s,r,\ell}$, posterior ambiguity is

$$A_{s,r,\ell}(S) = \sqrt{\boldsymbol{m}_{s,r,\ell}\boldsymbol{\Sigma}^{\mathrm{post}}_{s,r,S}\boldsymbol{m}^\top_{s,r,\ell}} \quad [\mathrm{mV}]. \tag{3}$$

PENDING–Stage 15

Robust ambiguity heatmaps
QRS, ST, and T; 64 configurations $\times$ 12 targets
color scale: A_{robust} (mV)

Fig. 1. Primary robust map. The frozen 64-row panel is drawn from the complete 255-configuration artifact. Observed targets remain visible for structural auditing; outcome analyses use only missing targets. Final pixels must come from the reviewed `figures_v3` artifact.

It is a predictive standard deviation under Eq. (1), conditional on the fitted representation and the frozen observation model.

2.2. Patient-bootstrap rank envelope

Each spatial model is refit in 2,000 patient-cluster bootstrap draws. With $Q^{\text{pat}}_{.975}$ denoting the empirical 97.5th percentile over those draws, the primary map is

$$A_{\text{robust}}(s, \ell, S) = \max_{r \in \{2,3,4,5\}} Q^{\text{pat}}_{.975}[A_{s,r,\ell}(S)]\,. \tag{4}$$

Thus no favorable rank can replace the prespecified family. The structural artifact contains all $2^8 - 1 = 255$ nonempty independent-channel configurations. Reconstruction experiments use an outcome-independent 64-configuration panel: all one- and two-channel sets plus a salted-hash sample at larger sizes. Its ordered membership and SHA-256 digest are frozen before outcomes. Figure 1 shows those same 64 rows so the displayed map and deep benchmark align.

The role of τ^2 is deliberately limited. It regularizes Gaussian conditioning and absorbs unrepresented variation at the observation interface; it is not presented as a calibrated sensor noise measurement. A single global choice prevents each target, segment, configuration, or rank from receiving a favorable post hoc value. In every bootstrap draw the patient blocks are sampled with replacement, all spatial ranks are refit, and Eq. (3) is recomputed with the same frozen τ^2. The percentile is formed within rank before the maximum in Eq. (4); bootstrap rows are never pooled as if rank were a random draw. Rank-deficient refits and their rejection fraction are retained in the audit, and insufficient valid refits stop release rather than silently narrowing the interval.

Diagnostics only. Let η be the target-row fraction outside the observed-row span. We retain $R_{\text{lower}} = \text{clip}\{1 - \max_r Q^{\text{pat}}_{.975}(\eta), 0, 1\}$, upper $\log_{10} \kappa$ summaries for target and global pseudoinverse amplification, and the point-estimate rank span $\max_r A_{s,r,\ell} - \min_r A_{s,r,\ell}$ in mV. These quantities audit geometry and rank sensitivity; they neither replace A_{robust} nor enter the primary incremental predictor.

3. LOCKED EVALUATION

3.1. Reconstructors, cells, and outcome

The common panel contains four primary methods: low-rank conditional mean, ridge regression, a mask-conditioned 1-D U-Net, and the pinned official ImputeECG implementation [7]. They receive the same patient partitions, 500-Hz delineated windows, whole-lead masks, reversible training normalization, and missing-target scorer on all 64 configurations. Predictions are returned to mV before evaluation. ECGrecover [6] is a strong pinned baseline for its supported single independent input lead. It is reported in a separate block and is not counted among the four common-panel methods. Missing source, command, data, or checkpoint artifacts are hard failures; public methods are not replaced by local surrogates.

Low-rank and ridge runs are deterministic. The U-Net, ImputeECG, and ECGrecover use model seeds 0–4. The shared scorer produces one missing-target outcome per patient, method, seed, segment, configuration, and target. The primary response is patient-level log-RMSE in mV on missing target samples. Other error summaries are supplementary. QRS, ST, and T are primary, while P-wave analysis is supplementary.

Population and cell construction. The primary PTB-XL population includes all diagnostic groups; a normal-rhythm subset is reserved for sensitivity analysis. Loaders enforce the canonical lead order, mV units, signs, 500-Hz rate, and patient identifiers before any model sees a record. A frozen DWT delineator [11] produces segment indices, and a record without evaluable primary windows is recorded with its exclusion reason. Whole leads, not isolated time samples, are hidden. A target already present in S is removed from the prediction endpoint, while its map value remains available for the structural audit in Fig. 1. Errors are accumulated over all eligible samples and records for a patient before the patient row is emitted. Consequently, a patient with more beats does not become multiple resampling units. The same construction code produces fold-8, fold-9, fold-10, and external outcome tables.

3.2. Fold discipline and incremental estimand

PTB-XL folds 1–7 fit spatial and reconstruction models, fold 8 contains tuning only, fold 9 is the meta-fit/calibration partition, and fold 10 is evaluated once [12]. In addition to τ^2, fold 8 selects one shared meta-model ridge penalty by leave-one-configuration-out (LOCO) prediction. Its neutral objective is the mean validation MSE of the simple and augmented meta-models. No fold-10 response can affect fitting or selection.

Both meta-models include categorical method, segment, and target effects. The simple numeric predictors are $|S|$, observed-matrix rank, $\log_{10}$ condition number, target RMS, and the largest observed–target correlation. The augmented model adds only A_{robust}. For every test configuration, the fold-9 fit excludes that configuration before predicting its fold-10 patient rows. The sole primary effect is

$$\Delta R^2 = R^2_{\text{augmented}} - R^2_{\text{simple}} \tag{5}$$

over the resulting LOCO predictions. Patient-cluster bootstrap uncertainty uses 2,000 draws; each draw also samples one available neural seed within each neural method. This nested seed operation prevents replicated checkpoints from masquerading as independent patients.

LOCO tests the intended generalization axis. A conventional random split of configuration–target rows could let the meta-model memorize a nearly identical observed set; here, every fold-10 configuration is predicted by models trained on fold 9 patients and other configurations. Numeric features are standardized within the fit and categorical levels are one-hot encoded with unseen levels tolerated. The fold-8 penalty grid is $\{0, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10, 10^2, 10^3\}$, with the smaller penalty breaking a tied validation criterion. The reported R^2 values are computed on the same retained patient rows, so ΔR^2 is a paired predictive comparison rather than the difference between separately filtered analyses. Method-specific point effects use those predictions only for the Stage 15 sign count; they do not create five additional headline hypotheses.

3.3. Two zero-transfer cohorts

Chapman–Shaoxing–Ningbo [13] and CPSC 2018 [14] each use a frozen, patient-hashed 60/20/20 manifest. The 60% partition is reserved for a secondary cohort-map sensitivity and the 20% development partition for audit/QC. Primary zero transfer reloads PTB-XL checkpoints and applies the fold-9 meta-fit, frozen τ^2, frozen ridge penalty, preprocessing, masks, and scorer to the untouched 20% test partition. It performs no cohort-specific fitting, adaptation, or threshold selection. Both external bundles must be complete for release; a secondary map fitted on an external 60% partition cannot substitute for either zero-transfer test.

The external target is performance transport, not re-estimation of spatial geometry. Each external patient row is therefore joined to the PTB-XL A_{robust} cell for its frozen segment, configuration, and target. A missing map cell, common-panel method, seed bundle, or test partition is a release error. Manifest audits attempt every authorized record

PENDING–Stage 15

Incremental predictive value ΔR^2

PTB-XL fold 10, Chapman test, CPSC 2018 test

patient-bootstrap 95% intervals

Fig. 2. Primary effect display for the untouched PTB-XL and zero-transfer test cohorts. Points and intervals are synchronized from the reviewed Stage 15 artifact; the zero line and cohort order are frozen.

and preserve the reason for any signal, length, unit, or delineation exclusion. The external 60% maps may quantify ranking shift as a secondary analysis, but they never supply the score used in Fig. 2. This prevents external test morphology from leaking into the proposed predictor while making cohort shift visible rather than treating successful file loading as evidence of transfer.

4. EVIDENCE GATE AND RESPONSIBLE SCOPE

Stage 15 is fail-closed. The automatic decision is PROCEED only when (i) the PTB-XL fold-10 lower 95% bootstrap bound for ΔR^2 is greater than zero; (ii) the corresponding lower bound is greater than zero in at least one of Chapman or CPSC 2018; and (iii) at least three of the four common-panel methods have positive point ΔR^2. Every other outcome is PIVOT. Either automatic outcome awaits user review for up to 24 hours; the claim macros and two final figures are released only after a valid human review signature.

Presentation contract. The main paper has two claim-bearing figures and one claim table. Figure 1 is purely structural and must contain all $64 \times 12 \times 3$ frozen cells. Figure 2 contains PTB-XL, Chapman, and CPSC 2018 in a fixed order, with the zero line and patient-bootstrap intervals. Table 1 is populated from the same reviewed decision artifact. Source tables, decision metadata, and rendered figures are hashed together; a manually typed value or manually substituted image fails the submission build. Point estimates and interval endpoints are reported together, and method-specific signs are shown even when they oppose the pooled effect. Until review, the visible placeholders are part of the paper's evidence boundary rather than estimates.

Falsification paths. The protocol distinguishes an informative negative result from a workflow failure. If the PTB-XL interval crosses zero, adding A_{robust} has not improved held-out prediction and the central claim is unsupported. If the pooled effect is positive but fewer than three common-panel

Table 1. Single claim table synchronized from the reviewed Stage 15 artifact.

Evidence item	Synchronized value
Reviewed primary claim	PENDING–STAGE 15
PTB-XL LOCO ΔR^2	PENDING–STAGE 15
Rank 2–5 envelope	PENDING–STAGE 15
Chapman and CPSC zero transfer	PENDING–STAGE 15
Four-method sign coverage	PENDING–STAGE 15
Patient/seed uncertainty	PENDING–STAGE 15

methods have positive point effects, the result is too method-specific for the proposed scope. If neither external lower bound is positive, transfer is unsupported even when the PTB-XL effect passes. A favorable raw association, rank-specific fit, external-train cohort map, or ECGrecover single-input result cannot override any of these outcomes. Incompatibility of a required official adapter, incomplete external test coverage, leakage, or failure to retain the required bootstrap draws blocks the gate rather than being recoded as a statistical null. This separation prevents implementation failures from being used selectively and prevents secondary evidence from rescuing a failed primary estimand.

Sensitivity hierarchy. Predeclared sensitivities cover raw 12-lead PCA, the PTB-XL normal-rhythm subset, another delineator, other error summaries, and maps fitted on each external 60% partition. Rank-resolved posterior ambiguity and the geometric diagnostics can explain a broad envelope but cannot nominate a replacement rank after fold 10. P-wave results remain supplementary because delineation completeness differs from the primary segments. These analyses are reported with their original signs and uncertainty after the primary decision; none changes the Stage 15 rule, predictor set, external test partitions, or title. Any analysis proposed after opening fold 10 is labeled exploratory in the ledger and is excluded from claim synchronization.

Artifact and leakage contract. Patient identities are disjoint across every PTB-XL role and across each external manifest partition. The evaluation loader freezes record membership, first-window boundaries, segment indices, and whole-lead masks before inference. Observed samples must be preserved pointwise, and only missing targets enter the response. Every claim-bearing prediction carries cohort and split hashes, the ordered configuration-panel hash, method and seed, source revision, environment, tuning artifact, and checkpoint hash. Failed records and bootstrap refits remain in audit files with reasons. The two figures are generated from source tables whose hashes are tied to the same Stage 15 decision; the manuscript never reads ad hoc result paths. Claim macros are produced by the synchronization script only, and a valid review signature is checked again during the five-page submission build. This contract makes a rerun traceable from the PDF cell back to patient-level predictions without exposing patient data in the paper.

Scope and limitations. The Gaussian prior and linear spatial representation are empirical approximations; learned reconstructors may use repeatable nonlinear or temporal associations absent from them. The frozen observation variance conflates residual model mismatch and measurement noise. Delineation, pathology prevalence, devices, placement, and acquisition noise can shift both maps and errors. Two external cohorts improve falsifiability but do not prove deployment validity. Reconstruction fidelity does not establish diagnostic equivalence or clinical safety, and R_{lower}, η, $\log_{10}\kappa$, and rank span are diagnostics, not guarantees. Null effects, reversals, failed seeds, exclusions under frozen rules, and adapter failures stay in the evidence ledger.

Data, anonymity, and AI disclosure. The submission is anonymous and uses only public, de-identified PTB-XL, Chapman, and CPSC 2018 data; it collects no new human-subject data. OpenAI Codex and Anthropic Claude Code assisted repository analysis, code/manuscript review, and editing. A bounded AutoResearchClaw v0.5.0 co-pilot probe was run with pinned ACP adapters, but Stage 01 failed before producing an artifact; no AutoResearchClaw research or review output was used. AI systems are not authors; a human remains responsible for protocol decisions, citation and source verification, execution, interpretation, every numerical value, and the submitted text. The tool/model/version ledger and venue-policy wording must be frozen before submission.

5. CONCLUSION

This study asks a deliberately narrow question: does a conservative Gaussian posterior ambiguity in mV improve out-of-sample prediction of patient reconstruction difficulty? The answer is reserved for one fold-9-to-fold-10 LOCO analysis, four common-panel methods, two zero-transfer cohorts, nested patient/seed uncertainty, and the hard Stage 15 thresholds. PENDING–STAGE 15

6. REFERENCES

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